Master's Project Proposal- StemMixRT VST

Griffin Barbieri, Alexander Lerch

***Abstract*—Pop producers are often confronted with poorly mixed 2-track sessions (vocals and combined instrumental) and full mixes (single combined track with vocals and instrumentals). If producers want to mix individual stems– i.e. percussion, bass, vocals, and other– they have access to only a few limited plugins capable of manipulating each source in real time. We propose a mixing and mastering VST plugin based on the HS-TasNet source separation model. We conceptualize it to give the user the following abilities:**

- **Create an arbitrary number of tracks each with user-defined stem combinations**
- **Add their own VSTs to each track**
- **Maximize audio quality by avoiding separation artifacts**

We hypothesize it will earn a higher rating on the Creativity Support Index [1]than the base StemGenRT [2].

***Index Terms*—Source Separation, VST Plugin, HCI, Music Production**

I. Research Statement

Commercial music producers often receive only combined instrumentals (drums, bass, and other instruments summed into one track) or full mixes (all instrumentals including vocals) to work with. When producers wish to improve those tracks, they have limited means of processing individual stems. We are interested in improving producer workflow in these contexts. We hypothesize that, by giving our plugin more functionality than basic channel functions, we could see a significant workflow improvement in 2-track mixing and mastering contexts as measured by CSI [1].

II. Motivation

If they wish to process individual stems with AFX, they have to either apply specific AFX discriminately against dynamic or timbral characteristics of each track, or use a third-party stem separation tool. Of those third-party tools, only a handful of them are VSTs that can operate in real-time without a time-intensive offline separation operation. Of those VSTs, all but iZotope's RX 12 Music Rebalance of them provide robust functionality besides channel buttons (gain, solo, and mute). iZotope's RX 12 Music Rebalance tool only allows users to put a handful of cleanup plugins (RX De-clicker, De-esser, etc.) We aim to offer producers a tool that goes beyond current industry limitations.

III. Related Work

A. Source Separation

Historical approaches to SS were based on DSP (filtering, panning, repetition-based), but current research focuses on ML. Spectrum estimation paradigms find the magnitude each instrument occupies in every FFT bin (e.g., GSEP [3]), and complex spectrum estimation paradigms find both the magnitude and phase of each instrument in every bin (e.g., [4]). Waveform estimation approaches typically encode audio into basis channel representations, estimate the mask of each stem across every dimension for every frame, and decode that back into audio of each separated stem [5].

TABLE I
BSSEval SDR (dB) for Source Separation Models

Model	Year	Mean SDR	Real-Time
Spleeter* [4]	2020	5.91*	N
Hybrid Demucs [7]	2021	7.64	N
BS-Roformer [6]	2024	9.80	N
HS-TasNet [5]	2024	4.65	Y (23 ms)

*Evaluated on non-HQ MUSDB18; others use MUSDB18-HQ.

One model commonly hailed as the SOTA is BS-Roformer [6], based on complex spectrum estimation. BS-Roformer implements a hierarchy of transformers within, and across, distinct frequency bands to estimate the spectral mask of each frame.

However, a model with a hybrid architecture [7] stood out as the best in SS for multiple years before tools like BS-Roformer, Hybrid Demucs. Two of the most prominent models today, HS-TasNet [5] and Hybrid Demucs [7], both use a combined waveform and spectrum estimation architecture. HS-TasNet operates in real time, whereas Hybrid Demucs and BS-Roformer operate offline with a slightly higher accuracy (see Table I). Despite advances in offline models, HS-TasNet is still the SOTA for real-time SS, and for that reason it is the ideal model to use for a RT SS plugin. The open source plugin StemGenRT [2] implements HS-TasNet with JUCE and ONNX, and it is from this plugin we will start our work.

B. Commercial Applications

SS tools are relatively commonplace. A variety of open-source SS models like Demucs and HS-TasNet means audio software companies can easily base new plugins on existing research. A large market for these tools exists. DJs are one market, as they often combine and remix the stems of one song with those from another. Commercial music producers are another audience, wherein producers may not have access to individual stem files. However, in informal conversations with Nashville producers we found that producers may not go out of their way to find a standalone stem splitter. These producers often rely on DAW session templates to expedite their mixing process and therefore resist opening up external programs.

Some Digital Audio Workstations also have built-in source separation tools, notably FL Studio 2025, Logic Pro 11, and Ableton Live 12. Other common DAWs such as ProTools,

TABLE II
LOCAL PROCESSING, REAL-TIME CAPABILITY, VST AVAILABILITY, MODEL DISCLOSURE, AND ADDITIONAL-EFFECTS SUPPORT BY TOOL

Tool	Locally-Run	Real-Time	VST/Plugin	Fully Proprietary	Internal AFX?
Moises [8]	N	N	Y	N	N
Spleeter [4]	Y	N	N	N	N
LANDR [9], [12]	N	N	Y	Y	N
SpectraLayers [11]	Y	N	Y	Y	N
LALAL.ai [13]	Y	Y	Y	Y	N
Peel Stems 2 [10]	Y	Y	Y	Y	N
Acon Digital Remix [14]	Y	Y	Y	Y	N
Gaudio Studio [15]	N	N	N	N	N
Traktor Pro 4 [16], [17]	Y	N	N	Y	N
VirtualDJ Stems 2.0 [18]	Y	Y	N	Y	N
Serato Stems [19]	Y	Y	N	Y	N
iZotope RX (Music Rebalance) [17]	Y	Y	Y	Y	Y

Reaper, and Cubase do not have such capabilities. In contrast, SS VSTs seemingly satisfy the need for an ”in-the-box” SS solution. Moises.ai [8], LANDR [9], Peel Stems 2 [10], and Steinberg SpectraLayers [11] are a few of the best stem splitters on the market. The models behind Spleeter [4] and Moises.ai [8] are relatively transparent–both publish regularly and Spleeter’s model is entirely open-source. These tools vary in their real-time capabilities, and whether they require internet connection. Table II shows the varying formats of each application.

Most of these tools operate outside of RT, which is adequate for certain use cases. For instance, DJs may prepare their sets in advance and have a pre-selected repertoire of tracks to remix. However, some do work in real time, such as LALAL.ai, zplane’s Peel Stems 2, and iZotope’s RX 12 Music Rebalance tool. The desktop app implementation of LALAL.ai can bounce up to 7 stem files offline and 10 when connected to LALAL.ai cloud models. Its VST plugin implementation is more limited, only allowing the user to choose a single stem per track to either isolate or remove. iZotope’s Music Rebalance plugin lets users to apply iZotope’s AFX on individual stems separated from a single stereo track. The VST also allows users to download individual stems as files. Peel Stems 2 by zplane also gives users basic mixing functions like muting and soloing for each stem.

One SS plugin, Acon Digital Remix, stands out as especially similar to our plugin. It allows users to route individual stems to audio tracks so they can apply their own AFX to each. However, processing a sum of multiple tracks requires complex routing and track grouping. By supporting VSTs within the plugin itself, and by allowing users to create any number of tracks with any combination of stems, a producer using our plugin can avoid convoluted routing scenarios and keep their mixing in one window.



Fig. 1. Acon Digital Remix.

IV. PROPOSED METHOD

A. Functionality

Though StemGenRT already implements HS-TasNet in a plugin, it does so only with the same per-stem functionality as most SS plugins (mute, solo, gain). We identify multiple pieces of functionality that would turn this tool from a standard SS plugin to one that gives users genuinely novel creative controls over their production. We aim to build everything in the JUCE framework for C++. We conceptualize the tool such that users can create as many tracks as they desire, each track being an arbitrary mixture of each stem that the user can specify. To reduce separation artifacts, we will constrain that logic such that there is always a bypassed version of the track playing. We also plan to be able to host other plugins within the plugin. We will utilize JUCE plugin hosting functionality to build this feature. Lastly, this tool must operate in RT for maximum user convenience, so we base our work off the existing StemGenRT plugin using the HS-TasNet model.

This plugin has much functionality similar to that of a DAW, but contained within one window. Besides basic mixing functions, users will be able to effectively create ”buses”, ”sends”, and ”returns” by creating effect chains and sending only certain stems into them. This will cut down on the friction currently present in that workflow, which involves making several duplicate tracks with a different stem ”soloed” inside as many duplicate SS plugins.

B. User Interface

Figure 3 shows a mock-up of our interface, featuring all functions the prototype will make available to users. We built this mock interface to be intuitive and seamless. Standard channel buttons and sliders are arranged compactly to make an efficient layout clearly discernible by producers.

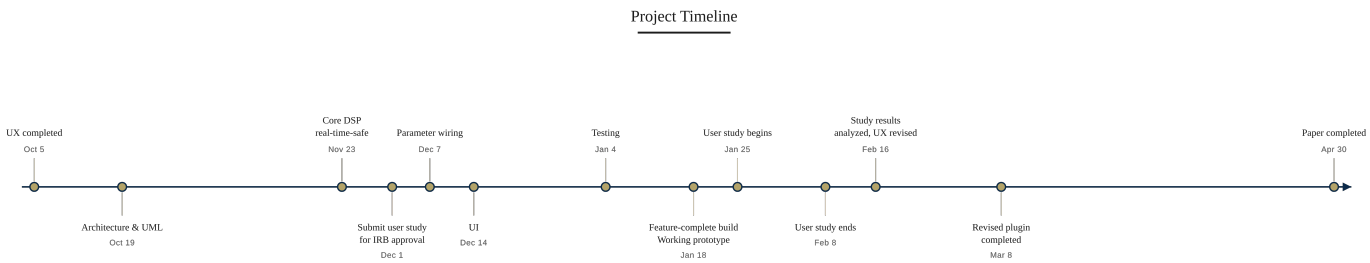


Fig. 2. Project timeline, from UX completion through the completed paper.

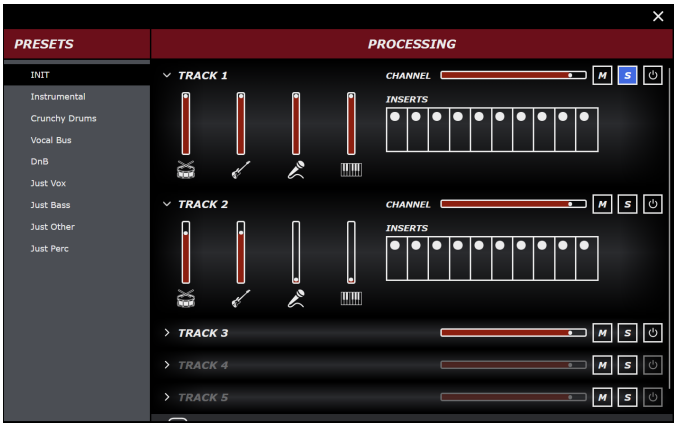


Fig. 3. Mock user interface of plugin.

C. Evaluation

We will conduct a formal user study after a prototype is developed to discover what other needs producers have with this tool. We currently plan to compare the CSI of the original StemGenRT, the DAW itself, and our StemMixRT plugin in a 2-track mixing and mastering scenario.

In addition to a CSI survey [1], we plan to gather written feedback from our users. We anticipate that users will want the following:

1) Higher-quality SS for pop genres.
2) Higher sample rate SS.
3) More robust performance across non-pop genres.
4) Basic native AFX suite.

We also wish to gauge their interest in an MCP for this tool so they can use LLMs to template their sessions.

V. Novelty

As previously stated, most SS plugins only have the basic mixing functions available (mute, solo, gain). iZotope's SS tool has some additional features, such as letting users add iZotope vocal clean-up plugins to each stem. Our tool will provide rarely-seen functionality in SS that will afford users real creative options, all the while improving producer workflow. Producers are often handed full mixes or 2-tracks they wish to mix again, either because the quality in the first place is poor or because they have creative ideas in mind. With current SS tools, producers would have to create a minimum of 4 duplicates of their master–for drums, bass, vocals, and other–if they wanted to process those stems separately. As previously mentioned, producers may not wish to open up external SS tools, and may generally avoid having several duplicate tracks with as many instances of a plugin. This tool would allow producers to bypass both those issues. By accelerating user workflow with integrated mixing features, it will allow them to render creative ideas more easily.

VI. Required Resources

We may utilize the Georgia Institute of Technology's PACE to experiment with implementations of HS-TasNet, and potentially fine-tuning the model on a non-pop dataset later in the duration of the project.

Additionally, we are seeking IRB approval for a user study to take place at the end of the prototyping phase in January. We aim to find a minimum of 10 producers to participate, and gauge whether this plugin facilitates the significant improvement in workflow that we predict it will have. We also aim to identify areas for feature improvement and avenues of future development.

Lastly, we will utilize Georgia Institute of Technology computers to perform our latency and stability testing on current hardware.

VII. Deliverables

We will deliver the following materials:

1) Prototype version of plugin (open source repository and VST).
2) Completed user study (n=10) examining producer engagement with tool.
3) Revised and updated version of plugin based on user feedback (open source repository and VST).

Our complete timeline is outlined in Figure 2. Priorities and deadlines may shift as the project progresses.

VIII. Conclusion

To sum up, we propose a RT SS plugin with novel creative features for pop producers. Where most other SS plugins only give users the most basic mixing functions, we propose a plugin that is capable of handling an arbitrary number of tracks each consisting of a user-defined stem mixture, and which allows users to add their own VSTs on those tracks. We plan to run a user study on our first prototype, and apply that feedback to the full release of an open-source SS plugin. This is a novel contribution to the SS plugin market, and after a user study, the revised plugin will be even more competitive with existing SS tools.

REFERENCES

[1] E. Cherry and C. Latulipe, "Quantifying the creativity support of digital tools through the creativity support index," *ACM Transactions on Computer-Human Interaction*, vol. 21, no. 4, Jun 2014.

[2] Sweet Spot Sound System, "StemGenRT: Real-time HS-TasNet stem separation (JUCE/ONNX)," GitHub repository, 2026. [Online]. Available: https://github.com/sweetspotsoundsystem/stemgen-rt

[3] S. Park and B. S. Chon, "Gsep: A robust vocal and accompaniment separation system using gated cbhg module and loudness normalization," 2021. [Online]. Available: https://arxiv.org/abs/2010.12139

[4] R. Hennequin, A. Khlif, F. Voituret, and M. Moussallam, "Spleeter: a fast and efficient music source separation tool with pre-trained models," *Journal of Open Source Software*, vol. 5, no. 50, p. 2154, 2020. [Online]. Available: https://doi.org/10.21105/joss.02154

[5] S. Venkatesh, A. Benilov, P. Coleman, and F. Roskam, "Real-time low-latency music source separation using hybrid spectrogram-tasnet," 2024. [Online]. Available: https://arxiv.org/abs/2402.17701

[6] W.-T. Lu, J.-C. Wang, Q. Kong, and Y.-N. Hung, "Music source separation with band-split rope transformer," 2023. [Online]. Available: https://arxiv.org/abs/2309.02612

[7] A. Défossez, "Hybrid spectrogram and waveform source separation," 2022. [Online]. Available: https://arxiv.org/abs/2111.03600

[8] Yun-Ning, Hung, I. Pereira, and F. Korzeniowski, "Moises-light: Resource-efficient band-split u-net for music source separation," 2025. [Online]. Available: https://arxiv.org/abs/2510.06785

[9] LANDR Audio, "LANDR stems," Product page, LANDR Audio Inc., 2024. [Online]. Available: https://www.landr.com/plugins/landr-stems

[10] zplane.development, "PEEL STEMS 2," Product page, 2026. [Online]. Available: https://products.zplane.de/products/peel-stems

[11] Steinberg Media Technologies GmbH, "SpectraLayers," Product page, 2026. [Online]. Available: https://www.steinberg.net/spectralayers/

[12] L. Miner, "Deep learning segmentation of audio using magnitude spectrogram," U.S. Patent Application 2021/0035590, 2021, assignee: AudioShake, Inc.; filed February 4, 2021. [Online]. Available: https://uspto.report/patent/app/20210035590

[13] LALAL.AI, "LALAL.AI: AI-powered stem splitter," Product website, 2026. [Online]. Available: https://www.lalal.ai/

[14] Acon Digital, "Remix," Product page, 2023. [Online]. Available: https://acondigital.com/products/remix

[15] Gaudio Lab, Inc., "Gaudio studio," Product website, 2026. [Online]. Available: https://gaudiolab.com/experience/source-separation

[16] Native Instruments, "TRAKTOR PRO 4," Press release / product page, 2024. [Online]. Available: https://www.native-instruments.com/en/press-area/djing/traktor-pro-4/

[17] iZotope, Inc., "RX: Music Rebalance," Product page, 2026. [Online]. Available: https://www.izotope.com/en/products/rx

[18] VirtualDJ, "VirtualDJ 2023: Stems 2.0," Vendor announcement, reported in DJ Mag, 2023. [Online]. Available: https://djmag.com/tech/virtualdj-2023-released-new-stem-separation-engine-studio-quality

[19] Serato, "Serato stems," Official announcement, 2023. [Online]. Available: https://the-drop.serato.com/announcements/a-new-era-for-djing-starts-now-with-serato-stems/